

Unit 4, Lesson 6: Finding Circumference

Benchmark

MA.7.GR.1.3 Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

Learning Target

- ☐ I can find the circumference of a circle.

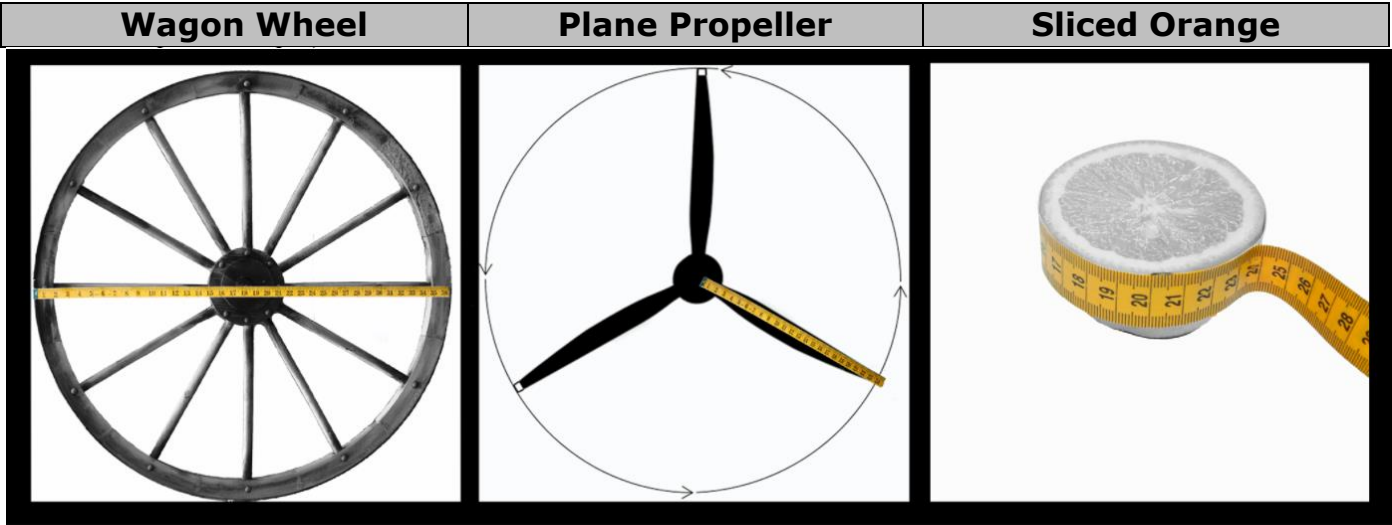
4.6.1 Warm-Up: Circular Objects

Three pictures of circular objects, with measurement tools, are shown. The measurement tool on each picture reads as follows:

Wagon wheel: 3 feet (ft)

Plane propeller: 24 inches (in)

Sliced Orange: 20 centimeters (cm)



1. For each object, what does the measurement represent?

Wagon Wheel	Plane Propeller	Sliced Orange
☐ radius ☐ diameter ☐ circumference	☐ radius ☐ diameter ☐ circumference	☐ radius ☐ diameter ☐ circumference

2. Choose one of the objects and explain why you think it would be helpful to know the measurement given.

4.6.2 Guided Instruction: Circumference

The relationship between the diameter, d , and the circumference, C , of a circle is proportional because the ratio of $C:d$ is always the same. This ratio, $\frac{C}{d}$, is called pi. The Greek symbol π is used to represent this value.

1. Rewrite the equation $\pi = \frac{C}{d}$ so that circumference, C , is written in terms of diameter, d .

$$C = \underline{\hspace{2cm}}.$$

2. Describe the relationship between the radius and the diameter of a circle.
3. Rewrite the equation from question 1 so that circumference, C , is written in terms of radius, r , by substituting how many radii make up the diameter, d .

$$C = \underline{\hspace{2cm}}.$$



Pi (π) is the symbol designating the ratio of the circumference of a circle to its diameter. It is an irrational number.

Common approximations for pi are 3.14, $\frac{22}{7}$ or $\frac{355}{113}$.

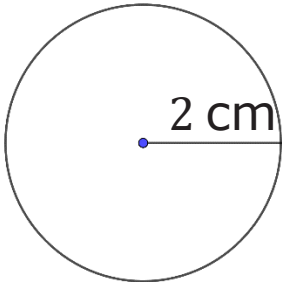
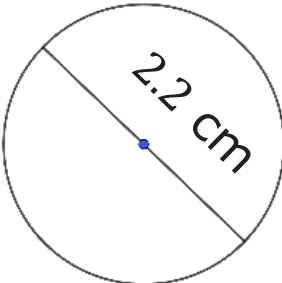
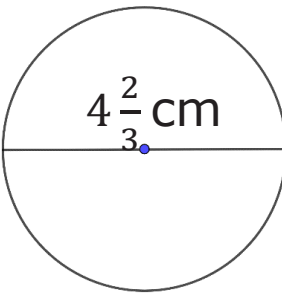
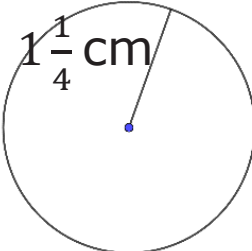
The symbol $\approx$ represents “is approximately.” It is used when a value has been rounded off or estimated.

The symbol π is used to represent the exact value of pi. Approximations of this value are often used when solving problems since pi is irrational and, therefore, has an infinite number of digits. When 3.14, $\frac{22}{7}$, or $\frac{355}{113}$ are used in place of π to solve problems, the answers are considered approximate values. Using the π button on a calculator will also result in an approximation because the result shown on the calculator does not include all the digits. Also, when using the value on the calculator, it is often rounded.

4. Complete the table to rewrite the circumference formulas using the common approximations for π .

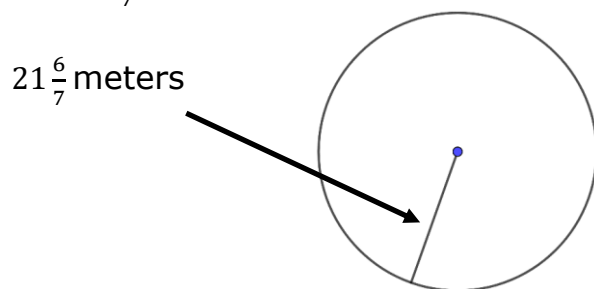
Approximate Value of pi (π)	$C = \pi d$	$C = 2\pi r$
$\pi \approx 3.14$	$C \approx 3.14d$	
$\pi \approx \frac{22}{7}$		
$\pi \approx \frac{355}{113}$		

5. Determine the circumference of each circle, both in terms of π and the approximation. For approximate values, use the approximation listed. Lengths are given in centimeters (cm). The circles are not drawn to scale.

	Terms of π	Approximate Value
		Use $\pi = \frac{355}{113}$.
		Use $\pi = 3.14$.
		Use $\pi = \frac{22}{7}$.
		Use $\pi = 3.14$.

4.6.3 Guided Practice: Error Analysis

1. Delanie used $\pi = \frac{22}{7}$ to find the circumference of the circle shown.



Delanie's work is shown.

Delanie's Work
$C = \pi d$
$C = \frac{22}{7} \left(21\frac{6}{7}\right)$
$C = \frac{22}{7} \left(\frac{153}{7}\right)$
$C = \frac{3366}{7}$
$C = 480\frac{6}{7} \text{ m}$

- a. Examine Delanie's work and write an explanation to her of any errors.
- b. What is the circumference of the circle?

4.6.4 Your Turn!

1. Stephanie is putting a fence around a circular garden.

If the garden has a diameter of 12.5 feet (ft), how many linear feet (ft) of fencing does Stephanie need to purchase? Round your answer to the nearest foot.



2. Rachel needs a new tube for her bicycle tire. The tire has a radius of 13 inches (in.). What is the circumference of the tire tube that Rachel should buy?

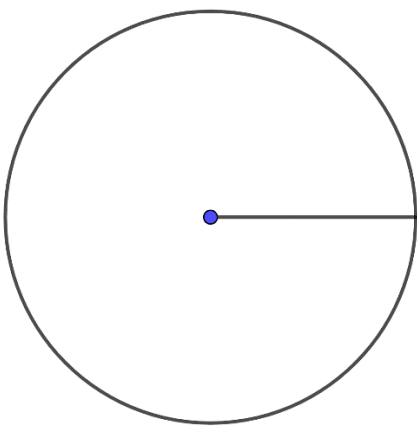


3. Natalia noticed that her hula-hoop is smaller than Felix's.
 - a. If Felix's hula-hoop has a diameter of 28 in., what is the circumference of his hula-hoop?
 - b. Natalia knows that her hula-hoop has a circumference of 81.64 in. Approximately how much larger is the circumference of Felix's hula-hoop than Natalia's?

4.6.5 Wrap-Up: Ticket Out the Door

Aziyia and Jamal are both finding the circumference of the circle shown. The radius of the circle is 7.3 inches (in.).

Some of Aziyia’s and Jamal’s work is shown.



Aziyia’s Work	Jamal’s Work
$C = 2\pi r$	$C = \pi d$
$C = (2)(\text{_____})r$	$C = \pi(2 \cdot \text{_____})$
$C = (2)(\text{_____})(\text{_____})$	$C = \text{_____}\pi \text{ in.}$
$C = 6.28(7.3)$	
$C = \text{_____} \text{ in.}$	

1. Complete each person’s work.
2. Explain which answer is more accurate.